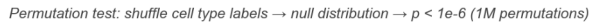


5. Quantify



A density plot showing the distribution of Procrustes distance. The x-axis is labeled 'Procrustes distance' and ranges from 60 to 120. The y-axis is labeled 'Density' and ranges from 0.000 to 0.200. A blue vertical line at approximately 61.15 represents the 'Observed' value. A grey histogram represents the distribution of 1,000,000 permutations, which is centered around 115. A text box in the lower left corner contains the following information: 'obs/null = 0.522', 'p < 1e-6', and 'n = 1,000,000 perms'.

Density

Procrustes distance

Within-lineage null

Global null

Observed (61.15)

Within-lineage: obs/null = 0.668 $p = 0.0001$

Global: obs/null = 0.522

This density plot shows the distribution of Procrustes distances. The x-axis ranges from 60 to 120. The y-axis represents density from 0.00 to 0.20. A blue vertical line at 61.15 represents the observed data. An orange histogram represents the within-lineage null model, centered around 92. A grey histogram represents the global null model, centered around 115. A text box provides statistical results for both null models.

CV = 0.0040
100/100 sig. at $\alpha = 0.01$
Gate: CV < 0.2 (PASS)

--- Mean = 61.18

Count (100 iterations)

Procrustes distance

Horizontal bar chart showing the LOOCV error / null ratio for various cancer types across six methods: Unsupervised, Immune, Epithelial, Stromal, Endothelial, and Metabolic. The x-axis ranges from 0.0 to 1.0. The y-axis lists cancer types. A vertical dashed line is at 0.5. The mean ratio is 0.4201.

Cancer Type	Unsupervised	Immune	Epithelial	Stromal	Endothelial	Metabolic
CRC: 1 cell	0.42	0.42	0.42	0.42	0.42	0.42
Non-classified cell	0.42	0.42	0.42	0.42	0.42	0.42
Luminal epithelial cell (mammary)	0.42	0.42	0.42	0.42	0.42	0.42
Apical cell	0.42	0.42	0.42	0.42	0.42	0.42
Endothelial cell	0.42	0.42	0.42	0.42	0.42	0.42
Enterochyle (large)	0.42	0.42	0.42	0.42	0.42	0.42
Intestine cell	0.42	0.42	0.42	0.42	0.42	0.42
B cell	0.42	0.42	0.42	0.42	0.42	0.42
Myeloid dendritic cell	0.42	0.42	0.42	0.42	0.42	0.42
Granulosa cell	0.42	0.42	0.42	0.42	0.42	0.42
Epithelial cell	0.42	0.42	0.42	0.42	0.42	0.42
Mesenchymal stem cell	0.42	0.42	0.42	0.42	0.42	0.42
Mesenchymal stromal cell	0.42	0.42	0.42	0.42	0.42	0.42
Myeloid cell	0.42	0.42	0.42	0.42	0.42	0.42
Macrophage	0.42	0.42	0.42	0.42	0.42	0.42
Fibroblast (small)	0.42	0.42	0.42	0.42	0.42	0.42
CDC17 cell	0.42	0.42	0.42	0.42	0.42	0.42
Smooth muscle	0.42	0.42	0.42	0.42	0.42	0.42
Myeloid leukocyte	0.42	0.42	0.42	0.42	0.42	0.42
Classical monocyte	0.42	0.42	0.42	0.42	0.42	0.42
Goblet cell (large)	0.42	0.42	0.42	0.42	0.42	0.42
Neutrophil	0.42	0.42	0.42	0.42	0.42	0.42
Stromal cell	0.42	0.42	0.42	0.42	0.42	0.42
Epithelial cell	0.42	0.42	0.42	0.42	0.42	0.42
Hematopoietic stem cell	0.42	0.42	0.42	0.42	0.42	0.42
Adipocyte	0.42	0.42	0.42	0.42	0.42	0.42
Hematopoietic cell	0.42	0.42	0.42	0.42	0.42	0.42

3526 ± 1.0
Mean = 0.4201